Some Applications of Trigonometry

The big idea

Surveyors find the height of a mountain or the width of a river using a single measured distance and an angle. This chapter applies tan, sin and cos to such **heights and distances** problems.

Key terms

- **Line of sight**: the line from the observer's eye to the object.
- **Angle of elevation**: the angle between the horizontal and the line of sight when you look **up**.
- **Angle of depression**: the angle between the horizontal and the line of sight when you look **down**.

ELEVATION = DEPRESSION

The angle of depression from A to B equals the angle of elevation from B to A, because they are alternate angles between parallel horizontal lines.

The method

1. Draw a neat figure. Mark the right angle, the given angle and the given length.
2. Choose the ratio that links the **known side** to the **unknown side**. Usually tan, because heights and horizontal distances are the two legs.
3. Use the exact values (tan 30° = 1/√3, tan 45° = 1, tan 60° = √3), then simplify.

A KITE FLIES AT 60 M ON A STRING INCLINED AT 60° TO THE GROUND. FIND THE STRING LENGTH.

$\sin 60° = 60 / L \Rightarrow L = 60 / (\sqrt{3}/2) = 120/\sqrt{3} =$ **$40\sqrt{3}$ m ≈ 69.3 m**.

THE ANGLE OF ELEVATION OF A TOWER'S TOP FROM A POINT 30 M AWAY IS 30°. FIND ITS HEIGHT.

$\tan 30° = h/30 \Rightarrow h = 30/\sqrt{3} =$ **$10\sqrt{3}$ m ≈ 17.3 m**.

Two-angle problems

When the observer moves towards the object, or there are two objects, you get **two right triangles**. Write a tan equation for each and solve them together.

FROM A POINT ON THE GROUND, THE ELEVATION OF A TOWER'S TOP IS 30°. WALKING 20 M CLOSER, IT BECOMES 60°. FIND THE HEIGHT.

Let height h, remaining distance x. $\tan 60° = h/x \Rightarrow h = \sqrt{3}x$. $\tan 30° = h/(x + 20) \Rightarrow \sqrt{3}h = x + 20$.

Substitute: $3x = x + 20 \Rightarrow x = 10$, so **$h = 10\sqrt{3}$ m ≈ 17.32 m**.

PUTTING THE ANGLE AT THE WRONG CORNER

The angle of elevation is at the **observer's eye**, between the horizontal and the line of sight. Draw it there.

Quick recap

- Elevation: looking up. Depression: looking down. They are equal for the same pair of points.
- tan links height and horizontal distance.
- Two angles ⇒ two triangles ⇒ two equations.

Practice questions

1. When the Sun's altitude changes from 30° to 60°, a tower's shadow gets
   (a) longer (b) shorter (c) unchanged (d) twice as long
2. A ladder 10 m long makes 60° with the ground. The height it reaches on the wall is
   (a) 5 m (b) 5√3 m (c) 10√3 m (d) 10 m
3. The angle of depression of a boat from a 50 m cliff is 45°. The boat is at a distance of
   (a) 25 m (b) 50 m (c) 50√3 m (d) 100 m
4. The angle of elevation of the top of a 15 m pole from a point is 60°. The distance of the point from the pole is
   (a) 5√3 m (b) 15√3 m (c) 15 m (d) 30 m
5. The angle of elevation from A to B is 35°. The angle of depression from B to A is
   (a) 55° (b) 35° (c) 145° (d) 90°

Answers: 1. b 2. b 3. b 4. a 5. b